

Challenge Name: phantom domain - 150 (Threat Intel L2)

Challenge Info

phantom domain
150

Description

33 Solves

Description

Your SOC intercepted a suspicious phishing email pointing to `https://secure-login[.]xyz`.

Task: Correlate them to find the attacker's hidden subdomain which contains the final flag.

File: [report.zip](#)

3/3 attempts

Flag

TL;DR

The phishing email pointed at `secure-login[.]xyz`. PDNS showed a few related hosts (e.g. `admin.secure-login.xyz`), but the decisive evidence was in the Certificate Transparency log (`ctlog.json`) where the certificate SANs included `internal-login.flagcorp.net`. The CTF organisers wrapped that hidden subdomain in the flag format:

```
SPL{internal-login.flagcorp.net}.
```

Reproduction (commands + relevant outputs)

Working directory:

```
(kali@kali)~/inf1n1ty/phantom
$ wget "https://bucket.secpen.org/phantom_domain.zip" -O report.zip
--2025-09-27 12:48:32-- https://bucket.secpen.org/phantom_domain.zip
Resolving bucket.secpen.org (bucket.secpen.org) ... 104.21.64.200, 172.67.155.212, 2606:4700:3031::6815:48ce, ...
Connecting to bucket.secpen.org (bucket.secpen.org)|104.21.64.200|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 468691 (458K) [application/x-zip-compressed]
Saving to: 'report.zip'

report.zip                               100%[=====] 457.71K 36.2KB/s in 13s
2025-09-27 12:48:45 (36.2 KB/s) - 'report.zip' saved [468691/468691]

(kali@kali)~/inf1n1ty/phantom
$ unzip report.zip
Archive: report.zip
  inflating: ctlog.json
  inflating: pdns.csv
  inflating: whois.txt
  inflating: auth.log
  inflating: banners.jsonl

(kali@kali)~/inf1n1ty/phantom
$ ls
auth.log  banners.jsonl  ctlog.json  pdns.csv  report.zip  whois.txt
```

Sanity check — no literal `flag{...}` sitting in the files:

```
(kali@kali)~/inf1n1ty/phantom
$ grep -Roi --binary-files=text "flag{[^\}]+\}" .

(kali@kali)~/inf1n1ty/phantom
$ grep -i "secure-login" pdns.csv
```

Look at PDNS for anything containing `secure-login`:

```
(kali@kali)-[~/1nf1n1ty/phantom]
$ grep -i "secure-login" pdns.csv
secure-login.co,2024-04-01,2025-01-12,150.49.114.34
secure-login.tech,2024-12-13,2025-09-22,125.239.63.234
secure-login.tech,2021-10-14,2025-06-17,93.232.39.146
secure-login-panel.io,2021-10-06,2025-04-01,67.155.70.42
secure-login-harsh.net,2021-08-15,2025-12-28,127.176.192.32
inteln-test-secure-login.org,2019-12-09,2024-01-29,15.136.206.107
pay-secure-login.tech,2023-04-05,2025-12-29,72.154.120.237
secure-login-vpn.io,2020-11-14,2025-09-24,162.37.164.14
secure-login.net,2024-01-31,2024-06-08,8.104.61.144
secure-login-news.info,2021-12-21,2025-05-16,29.1.107.184
secure-login-admin.org,2018-07-13,2025-10-31,222.55.145.244
harsh-secure-login.com,2020-03-13,2025-02-11,54.80.60.222
login-secure-login.com,2019-03-24,2024-05-18,200.134.11.102
secure-login.org,2018-06-28,2025-06-10,172.244.147.126
secure-login.tech,2020-06-15,2025-03-23,28.123.144.65
harsh-secure-login.io,2025-07-24,2024-01-15,152.25.86.222
secure-login.tech,2021-02-27,2024-11-08,115.224.1.48
staging-secure-login.net,2025-10-18,2025-10-25,178.197.236.112
secure-login.xyz,2025-07-11,2025-08-01,45.77.123.10
admin-secure-login.xyz,2025-08-03,2025-08-03,45.77.123.10
cdn.secure-login.xyz,2025-08-03,2025-08-03,45.77.123.11
```

Interpretation: attacker used `secure-login.xyz` variants — plausible phishing infrastructure, but not the final flag.

Scan banners for admin/login/internal-looking hosts (quick heuristic):

```
(kali@kali)-[~/1nf1n1ty/phantom]
$ grep -i "login|admin|internal" banners.jsonl | head
{"domain": "data-admin-pay.net", "http_title": "Site under construction", "server": "cloudflare", "favicon_md5": "bd83069c4b2b40b7", "ip": "143.253.142.16"}
{"domain": "staging.co", "http_title": "Login - Secure Portal", "server": "Apache/2.4.29 (Ubuntu)", "favicon_md5": "fa80c332c80b40fa", "ip": "145.52.150.6"}
{"domain": "panel.org", "http_title": "Login - Secure Portal", "server": "lighttpd/1.4.45", "favicon_md5": "80c3d7143fa4a5ee", "ip": "6.148.15.209"}
{"domain": "admin.xyz", "http_title": "OpenVPN Status", "server": "lighttpd/1.4.45", "favicon_md5": "5f280ff027788607", "ip": "32.93.108.150"}
{"domain": "shop-admin-shop.net", "http_title": "Site under construction", "server": "lighttpd/1.4.45", "favicon_md5": "cbcfde7bc81641bd", "ip": "70.40.226.70"}
{"domain": "admin-shop.net", "http_title": "Apache2 Ubuntu Default Page", "server": "cloudflare", "favicon_md5": "76d7122b08781f72", "ip": "44.80.110.156"}
{"domain": "go-data-go.xyz", "http_title": "Login - Secure Portal", "server": "cloudflare", "favicon_md5": "15d0c9d78a2499b6", "ip": "97.80.171.84"}
{"domain": "login-intel-login.io", "http_title": "Site under construction", "server": "nginx/1.14.0", "favicon_md5": "0a37beab0725d8ef", "ip": "110.229.162.12"}
{"domain": "cdn-admin.info", "http_title": "Site under construction", "server": "PleskOnyx v18.0.23 (Ubuntu)", "favicon_md5": "6ca8a53afd988bf8", "ip": "176.9.72.85"}
{"domain": "admin-shop-intel.tech", "http_title": "502 Bad Gateway", "server": "cloudflare", "favicon_md5": "ee9364ebe7966f53", "ip": "87.106.6.170"}
```

Mostly noisy; nothing that immediately reveals the final flag.

Pivot to Certificate Transparency logs (the goldmine):

```
(kali@kali)-[~/1nf1n1ty/phantom]
$ jq -r '.[.] | .san? // empty' ctlog.json | grep -i flagcorp -n
6300: "internal-login.flagcorp.net"

(kali@kali)-[~/1nf1n1ty/phantom]
$ grep -i "flagcorp" ctlog.json -n
20265: "internal-login.flagcorp.net"
```

This shows `internal-login.flagcorp.net` was registered as a SAN in a certificate observed in CT logs. That is the “hidden” subdomain the challenge asked us to find.

Conclusion / Reasoning

- The phishing pointer in the email (`secure-login[.]xyz`) leads to PDNS entries showing subdomains on that domain — plausible decoys.
- CT logs revealed that the same actor requested certificates covering other domains (SANs). One SAN was `internal-login.flagcorp.net` — an internal-sounding hostname not directly visible from the phishing URL.
- The organisers used that SAN as the hidden subdomain (the flag), formatted for the CTF as `SPL{internal-login.flagcorp.net}`.

Evidence (copy-paste)

From `pdns.csv`:

```
secure-login.xyz,2025-07-11,2025-08-01,45.77.123.10
admin.secure-login.xyz,2025-08-03,2025-08-03,45.77.123.10
cdn.secure-login.xyz,2025-08-03,2025-08-03,45.77.123.11
```

From `ctlog.json` (SAN excerpt):

```
"san": [
  "panel-login.com",
  "vpn.panel-login.com",
  "internal-login.flagcorp.net"
]
```

Final

Flag: `SPL{internal-login.flagcorp.net}`

~By: 1nfln1ty